

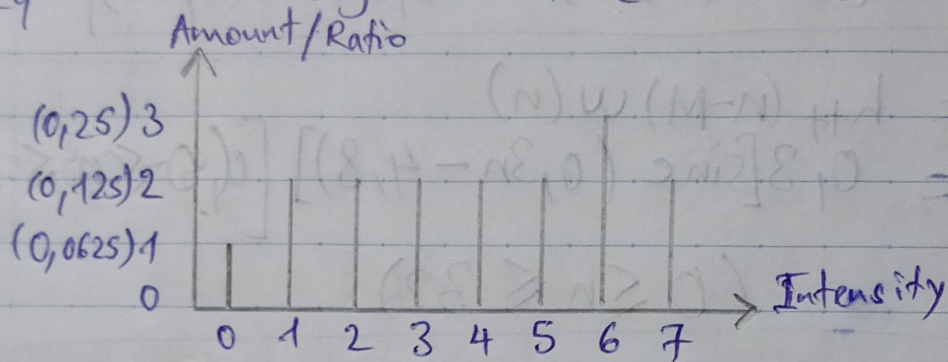
Digital Image Processing

Labwork 1

A1/

Intensity	0	1	2	3	4	5	6	7
Amount	1	2	2	2	2	2	3	2
Ratio	0,0625	0,125	0,125	0,125	0,125	0,125	0,28	0,125
CDF	1	3	5	7	9	11	14	16
N.CDF	0,0625	0,1875	0,3125	0,4375	0,5625	0,6875	0,875	1
New intensity (N.CDF * L)	0	1	2	3	4	5	6	7

→ equalized histogram is the same as original histogram



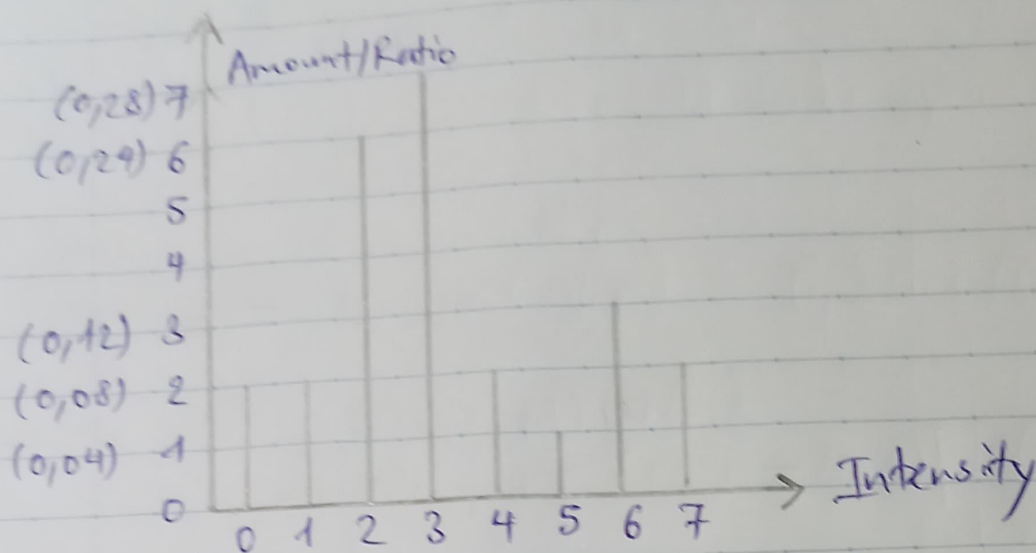
Median pixel value : 4

$$\rightarrow B = I_{\text{thresholding}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$I_{\text{negative}} = \begin{bmatrix} 7 & 6 & 6 & 5 \\ 5 & 4 & 4 & 3 \\ 3 & 2 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{bmatrix}$$

A2/

Intensity	0	1	2	3	4	5	6	7
Amount	2	2	6	7	2	1	3	2
Ratio	0,08	0,08	0,24	0,28	0,08	0,04	0,12	0,08



Mode: 3

$$\rightarrow B = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$B(x,y) = 1$: 8 pixels

$B(x,y) = 0$: 17 pixels

$k=2$ would be a better threshold, since the difference between the amounts of pixels in the foreground and the background would be smaller.